

Laporan Modul 10 — Security Hardening, Firewall & Audit Dasar

Workshop Administrasi Jaringan · PENS — Teknik Informatika dan Komputer Kelompok X
= 2 · Tanggal: 2026-06-14 · OS: Debian 13.4 (trixie)

1. Tujuan

Mengamankan server Linux dasar, mengonfigurasi firewall (nftables), melakukan audit service & port, dan mendeteksi brute-force SSH (fail2ban).

2. Topologi



3. Package

nftables, fail2ban, nmap, net-tools, curl, vim terpasang di srv01. nftables v1.1.3 · Fail2Ban v1.1.0 · Nmap 7.95.

4. Firewall (nftables)

File /etc/nftables.conf. Chain input policy **drop**, hanya membuka port layanan lab. Chain forward policy **drop** + izin routing antar subnet (fungsi router Modul 4). output accept. Stateful + log paket ditolak.

Port	Layanan	Status
22/tcp	SSH	open
53 tcp+udp	DNS	open
67/udp	DHCP	open
80, 443/tcp	HTTP/HTTPS	open
ICMP	ping	accept

Apply: nft -f /etc/nftables.conf ; systemctl enable nftables (persist boot).

Penting: ruleset modul (hanya 22/80/ICMP) sengaja diperluas. srv01 menjalankan DNS, DHCP, HTTPS, dan routing — memakai ruleset minimal modul akan mematikan seluruh layanan lab. Lihat `docs/hardening_policy_v1.md`.

5. fail2ban

Jail [sshd]: `maxretry=3`, `findtime=600`, `bantime=600`, backend `systemd`. File `/etc/fail2ban/jail.local`.

6. Pengujian

a. Audit port (srv01)

`ss -tulnp` → layanan terdaftar: 22, 53, 67, 80, 443 (sesuai kebijakan). Evidence: `evidence/audit_ports.txt`.

b. Verifikasi layanan tetap hidup (dari cli01)

DNS `dig app.lab2.local @10.20.2.1` → 10.20.2.1; HTTP 200; HTTPS 200; ping OK; SSH 22 open; DHCP lease aktif (10.20.2.10). Firewall tidak memutus layanan.

c. Scan nmap (cli01 → srv01)

22/53/80/443 = open; 3306/8080/9000 = **filtered** (di-drop policy). Membuktikan firewall menyaring port yang tidak diizinkan. Evidence: `evidence/nmap_scan.txt`.

d. Brute-force SSH

4x login gagal dari cli01 → fail2ban mendeteksi (Total failed: 4) dan **ban IP 10.20.2.10** (Total banned: 1). Setelah bukti diambil, IP di-unban agar klien tetap dipakai. Evidence: `evidence/fail2ban_status.txt`.

7. Penjelasan port yang terbuka

- 22 SSH: administrasi (publickey-only, hasil hardening Modul 1).
- 53 DNS: zona authoritative lab2.local + reverse (Modul 5).
- 67 DHCP: distribusi IP klien LAN-B (Modul 5).
- 80/443 HTTP/HTTPS: web server + reverse proxy TLS internal (Modul 6).
- Port lain (mis. 3306 MariaDB, 9000 backend) ditolak/tidak diekspos → backend python hanya bind 127.0.0.1, DB ada di host lain (adm01).

8. Bukti Screenshot

[1] nftables aktif (policy drop) + fail2ban (ban brute-force) — srv01

```
== MODUL 10 FIREWALL & HARDENING - srv01 ==
nftables: active
(enabled: enabled
)
fail2ban: active
--- chain input (policy drop) ---
type filter hook input priority filter; policy drop;
ct state established,related accept
iif "lo" accept
ip protocol icmp accept
ip6 nexthdr ipv6-icmp accept
tcp dport 22 accept
tcp dport { 80, 443 } accept
tcp dport 53 accept
udp dport 53 accept
udp dport 67 accept
--- fail2ban sshd ---
| - Total failed: 4
| - Total banned: 1
| - Banned IP list:
fishir@srv01:~$ _
```

[2] Audit nmap dari cli01 — port allowed=open, lainnya=filtered

```
== MODUL 10 AUDIT - nmap dari cli01 (10.20.2.10) -> srv01 (10.20.2.1) ==
Starting Nmap 7.95 ( https://nmap.org ) at 2025-06-14 00:11 WIB
Nmap scan report for app.lab2.local (10.20.2.1)
Host is up (0.00068s latency).

PORT      STATE SERVICE
22/tcp    open  ssh
53/tcp    open  domain
80/tcp    open  http
443/tcp   open  https
3306/tcp  filtered mysql
8080/tcp  filtered http-proxy
9000/tcp  filtered cslistener

Nmap done: 1 IP address (1 host up) scanned in 1.62 seconds
fishir@cli01:~$ _
```

9. Kesimpulan

Firewall nftables default-deny + fail2ban memberi hardening dan proteksi brute-force, sementara seluruh layanan lab (DNS, DHCP, web, routing) tetap berfungsi. Audit `ss` dan `nmap` mengonfirmasi hanya port yang diizinkan yang terbuka. *Server yang berjalan ≠ server yang aman* — keamanan butuh kontrol akses eksplisit, bukan sekadar layanan hidup.

10. Penyimpangan

- OS Debian 13 (bukan 12); fail2ban backend `systemd` (`journald`).
- Ruleset diperluas dari modul untuk menjaga DNS/DHCP/HTTPS/routing.
- nmap dijalankan dari cli01 (modul menyebut srv01) agar benar-benar menguji firewall dari sisi klien eksternal.